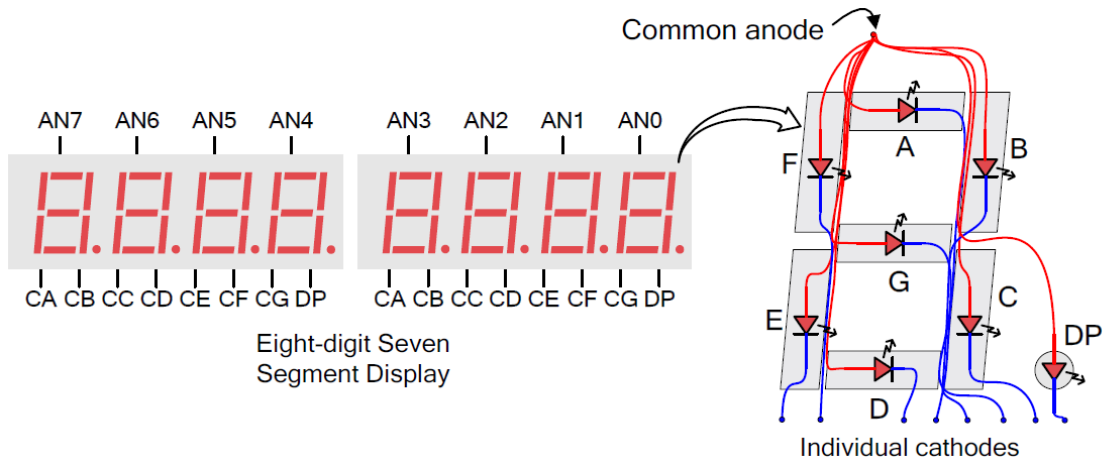


Lab4. 7-segment를 사용한 숫자 2자리 표시

다음과 같은 7-segment를 사용하여 lab1에서의 덧셈 결과를 2자리 16진수로 나타내기로 한다.



예를 들어 1을 나타낼 때는 b, c 만 ‘0’이고 나머지 LED는 ‘1’이다. 나머지 숫자는 다음 표를 참고하기 바람.

	a	b	c	d	e	f	g	dp	An7	An6	~	An0
0	0	0	0	0	0	0	1	1	0	1	1	1
1	1	0	0	1	1	1	1	1	0	1	1	1
2	0	0	1	0	0	1	0	1	0	1	1	1
3	0	0	0	0	1	1	0	1	0	1	1	1
4	1	0	0	1	1	0	0	1	0	1	1	1
5	0	1	0	0	1	0	0	1	0	1	1	1
6	0	1	0	0	0	0	0	1	0	1	1	1
7	0	0	0	1	1	0	1	1	0	1	1	1
8	0	0	0	0	0	0	0	1	0	1	1	1
9	0	0	0	1	1	0	0	1	0	1	1	1
a	0	0	0	1	0	0	0	1	0	1	1	1
b	1	1	0	0	0	0	0	1	0	1	1	1
c	0	1	1	0	0	0	1	1	0	1	1	1
d	1	0	0	0	0	1	0	1	0	1	1	1
e	0	1	1	0	0	0	0	1	0	1	1	1
f	0	1	1	1	0	0	0	1	0	1	1	1

twoseg.v 소스파일

```
module twoseg(clk, rst, a,b,y
    );
input clk, rst;
input [3:0] a,b;
output [15:0] y;

reg [4:0] sum;
reg [3:0] fbit;
reg [15:0] y;
reg [17:0] count;

reg s, ns;
localparam s0=0, s1=1;

always @ (a,b)
begin
    sum = a + b;
end

always@ (posedge clk)
begin
    s <= ns;
    if (rst == 1 || count >= 10**5) // simulation 10
        count <= 0;
    else
        count <= count + 1;
end

always@(rst, s, count, sum)
begin
    ns = s;

    case (s)
```

```

s0: begin
    y[8:0] = 9'b110111111;
    fbit = sum[3:0];
    ns = s0;
    if (count == 9)
        ns = s1;
end
default: begin
    y[8:0] = 9'b101111111;
    fbit = sum[4];
    ns = s1;
    if (count == 9)
        ns = s0;
end
endcase

if (rst == 1) begin
    ns = s0;
end

end

always@(fbit) begin
    case (fbit)
        0: y[15:9] = 7'b0000001;
        1: y[15:9] = 7'b1001111;
        2: y[15:9] = 7'b0010010;
        3: y[15:9] = 7'b0000110;
        4: y[15:9] = 7'b1001100;
        5: y[15:9] = 7'b0100100;
        6: y[15:9] = 7'b0100000;
        7: y[15:9] = 7'b0001101;
        8: y[15:9] = 7'b0000000;
        9: y[15:9] = 7'b0001100;
        10: y[15:9] = 7'b0001000;
        11: y[15:9] = 7'b1100000;
        12: y[15:9] = 7'b0110001;
    endcase
end

```

```

    13: y[15:9] = 7'b1000010;
    14: y[15:9] = 7'b0110000;
    15: y[15:9] = 7'b0111000;
    default: y[15:9] = 7'b1111111;
    endcase
end

always@ (posedge clk)
begin
    if (count == 8) begin
        if (s == s0)
            $display("s0: %d + %d ==> %b", a, b, y);
        else
            $display("s1: %d + %d ==> %b", a, b, y);
        end
    end
end
endmodule

```

테스트벤치 파일 **tb.v**

```
module tb;
    // Inputs
    reg [3:0] a;
    reg [3:0] b;

    // Outputs
    wire [15:0] y;

    // for loop
    reg [4:0] i, j;

    reg clk, rst;

    // Instantiate the Unit Under Test (UUT)
    twoseg uut (
        .a(a),
        .b(b),
        .y(y),
        .clk(clk),
        .rst(rst)
    );

    // clock generation
    initial begin
        clk = 0;
        forever #5 clk = ~clk;
    end

    // reset
    initial begin
        rst = 0;
        #100 rst = 1;
        #100 rst = 0;
    end
endmodule
```

```

end

initial begin
    // Initialize Inputs
    a = 0;
    b = 0;

    for (i=0; i<= 15; i=i+1) begin
        for (j=0; j<=15; j=j+1) begin
            #100; // $display("%d + %d ==> %b", a, b, y);
            #100 b = b + 1;
        end
        a = a + 1;
    end
end

endmodule

```

XDC 파일 twoseg.xdc

```
set_property PACKAGE_PIN E3 [get_ports clk]
set_property IOSTANDARD LVCMOS33 [get_ports clk]
create_clock -add -name sys_clk_pin -period 10.00 -waveform {0 5} [get_ports clk]
```

```
set_property PACKAGE_PIN N17 [get_ports rst]
set_property IOSTANDARD LVCMOS33 [get_ports rst]
```

```
set_property PACKAGE_PIN J15 [get_ports {a[0]}]
set_property IOSTANDARD LVCMOS33 [get_ports {a[0]}]
```

```
set_property PACKAGE_PIN L16 [get_ports {a[1]}]
set_property IOSTANDARD LVCMOS33 [get_ports {a[1]}]
```

```
set_property PACKAGE_PIN M13 [get_ports {a[2]}]
set_property IOSTANDARD LVCMOS33 [get_ports {a[2]}]
```

```
set_property PACKAGE_PIN R15 [get_ports {a[3]}]
set_property IOSTANDARD LVCMOS33 [get_ports {a[3]}]
```

```
set_property PACKAGE_PIN R17 [get_ports {b[0]}]
set_property IOSTANDARD LVCMOS33 [get_ports {b[0]}]
```

```
set_property PACKAGE_PIN T18 [get_ports {b[1]}]
set_property IOSTANDARD LVCMOS33 [get_ports {b[1]}]
```

```
set_property PACKAGE_PIN U18 [get_ports {b[2]}]
set_property IOSTANDARD LVCMOS33 [get_ports {b[2]}]
```

```
set_property PACKAGE_PIN R13 [get_ports {b[3]}]
set_property IOSTANDARD LVCMOS33 [get_ports {b[3]}]
```

```
#a
```

```
set_property PACKAGE_PIN T10 [get_ports {y[15]}]
set_property IOSTANDARD LVCMOS33 [get_ports {y[15]}]
```

```
#b
```

```

set_property PACKAGE_PIN R10 [get_ports {y[14]}]
set_property IOSTANDARD LVCMOS33 [get_ports {y[14]}]
#c
set_property PACKAGE_PIN K16 [get_ports {y[13]}]
set_property IOSTANDARD LVCMOS33 [get_ports {y[13]}]
#d
set_property PACKAGE_PIN K13 [get_ports {y[12]}]
set_property IOSTANDARD LVCMOS33 [get_ports {y[12]}]
#e
set_property PACKAGE_PIN P15 [get_ports {y[11]}]
set_property IOSTANDARD LVCMOS33 [get_ports {y[11]}]
#f
set_property PACKAGE_PIN T11 [get_ports {y[10]}]
set_property IOSTANDARD LVCMOS33 [get_ports {y[10]}]
#g
set_property PACKAGE_PIN L18 [get_ports {y[9]}]
set_property IOSTANDARD LVCMOS33 [get_ports {y[9]}]
#dp
set_property PACKAGE_PIN H15 [get_ports {y[8]}]
set_property IOSTANDARD LVCMOS33 [get_ports {y[8]}]
#an7
set_property PACKAGE_PIN U13 [get_ports {y[7]}]
set_property IOSTANDARD LVCMOS33 [get_ports {y[7]}]
#an6
set_property PACKAGE_PIN K2 [get_ports {y[6]}]
set_property IOSTANDARD LVCMOS33 [get_ports {y[6]}]
#an5
set_property PACKAGE_PIN T14 [get_ports {y[5]}]
set_property IOSTANDARD LVCMOS33 [get_ports {y[5]}]
#an4
set_property PACKAGE_PIN P14 [get_ports {y[4]}]
set_property IOSTANDARD LVCMOS33 [get_ports {y[4]}]
#an3
set_property PACKAGE_PIN J14 [get_ports {y[3]}]
set_property IOSTANDARD LVCMOS33 [get_ports {y[3]}]
#an2

```



```
set_property PACKAGE_PIN T9 [get_ports {y[2]}]
set_property IOSTANDARD LVCMOS33 [get_ports {y[2]}]
#an1
set_property PACKAGE_PIN J18 [get_ports {y[1]}]
set_property IOSTANDARD LVCMOS33 [get_ports {y[1]}]
#an0
set_property PACKAGE_PIN J17 [get_ports {y[0]}]
set_property IOSTANDARD LVCMOS33 [get_ports {y[0]}]
```

[도전문제 1] 입력 **a, b**를 각각 **6비트**로 변경하고, 덧셈결과를 **8진수**로 **7-segment 3**개에 나타내도록 코드를 수정하라

[도전문제 2] 입력 **a, b**를 각각 **6비트**로 변경하고, 덧셈결과를 **십진수**로 **7-segment 3개**에 나타내도록 코드를 수정하라

참고> 정수연산 $345 \% 10 = 5$, $(345/10)\%10 = 4$, $345/100 = 3$